Flight Geometry Is the Schedule: Pre-Planned Ground Cooperation for Single-Pass UAV Query Delivery to Heterogeneous Edge-Camera Networks

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***Abstract*—To find a missing person quickly, a search query—a set of reference images, hundreds of kilobytes—must reach the edge cameras of a large, heterogeneous IoT network that has no broadband backhaul. A fixed-wing UAV can broadcast the query over IEEE 802.15.4, but it passes each cluster once, and how far its signal reaches varies from about 1 km over open land to below 100 m under forest canopy or among tall buildings. We observe that in a single-pass broadcast *the flight geometry is the transmission schedule*: the base station knows, before take-off, which segment of the stream is emitted above which cell. We exploit it on both sides of the link. *LM-Tour* plans the flight through a weighted, well-spread set of landmark AI nodes, choosing their spacing so that the ground can serve everything in between. *River*, a pre-planned cell-level transport, moves each segment from the cells under the path, through bank cells and a ring of helper cells, to the few AI-capable cameras that run the matcher. It needs no route discovery, splits long contended paths into short parallel hops, works unchanged with fountain-coded packets, and reuses the same routes in reverse to pull event frames from legacy cameras that cannot process images themselves. In ns-3 with a full 802.15.4 MAC, River completes the last critical camera 26% earlier with 12% fewer frames than direct pulling, its advantage grows with the number of critical cameras, and under short air-to-ground range ($r \leq 200$ m) it completes all critical cameras while fountain coding with in-cell gathering leaves X% incomplete. The full mission is 1.9–2.8× faster than rotary-wing alternatives.**

***Index Terms*—UAV, single-pass broadcast, IEEE 802.15.4, cooperative dissemination, edge AI, search and rescue.**

I. Introduction

UAVs have moved from demonstrations to daily operations at remarkable speed: they inspect power lines, map crops, deliver medicine and extend cellular coverage after disasters [2]. Most of these uses make an existing service faster or cheaper—delivering goods, relaying data, extending a network. UAVs also enable a different kind of task: *operations* that did not exist before because no infrastructure could support them, typically urgent, one-off missions in places where the network is weak.

Searching for a missing person is such a mission. A missing child or a lost hiker has often walked past many cameras installed by different owners at different times—municipalities, shops, park services, road operators. A few of them, or an edge box attached to them, can run a neural matcher; most are legacy devices that only record. None knows whom to look for. Where an AI camera has broadband (Wi-Fi, fibre, cellular), the query can simply be sent to it; our concern is the large remainder in rural towns, mountain resorts, parks and along rural roads, where devices are linked only by low-power radios and footage cannot be uploaded.

We therefore push a compact *query* down instead of pulling footage up. If all AI nodes ran the same model, a few kilobytes of feature vectors would suffice. In a network built by many owners they do not, and vectors of different models are not comparable [19]; the query must then carry reference images, so that each node embeds them with its own model. Its size is a scenario parameter—from a face-only query of tens of kilobytes to several hundred kilobytes for faces, full body and clothing; we use $L = 12$ images, $Q = 4{,}036$ packets (≈400 kB) as the nominal case and treat the vector query as a baseline. Where no backhaul exists, a fixed-wing UAV carries the query.

Challenge. A fixed-wing UAV flies fast (50 m/s) but cannot hover; it crosses each cluster once. One copy of the query takes $Q\Delta \approx 2$ km of flight. If the air-to-ground (A2G) range r is about 1 km (open land), nodes far from the path still receive many packets, but scattered, so few complete an image. If r is below 100–200 m (forest canopy, tall buildings), a cell hears only the ≈200–400 packets emitted while the UAV is overhead—a single image at best. Either way, the AI nodes (*critical cells*), which are rarely on the path, cannot assemble the query alone.

Key insight. Packet s is emitted at $\gamma(d_0 + s\Delta)$ on a path γ that the base station (BS) designs. Hence before take-off the BS knows which segment of the stream lands on which cell, and can plan both the path and the ground cooperation. Existing UAV multicast designs require every receiver to obtain the file directly from the UAV [20]; dissemination protocols such as Deluge [21] discover data at run time and target every node; fountain codes [22] relax which packets are needed but not where they are.

Contributions.

1) We formulate single-pass query delivery to a heterogeneous camera network around a *reception map* predicted from the planned trajectory, with a lightweight model of what partial data costs the search (Section III).
2) *River*: a pre-planned, bidirectional cell-level transport—Spine → Bank → Ring → critical cell for the query, and per-critical-cell trees for pulling event frames from legacy cameras—compatible with fountain coding (Section IV).

3) LM-TOUR, a flight design that covers the AI nodes with a weighted R-net, flies a Dubins tour through their neighbourhoods and inserts further nodes by value per extra metre; R trades tour length ($\propto 1/R$) against River distance ($\propto R$) (Section V).
4) An ns-3 evaluation with a full 802.15.4 MAC across A2G regimes, against direct pulling and fountain coding with in-cell gathering (Section VI).

II. RELATED WORK

UAV-enabled communication. UAV trajectories are routinely optimised jointly with communication [1], [2], [3], including fixed-wing energy models [4]. Closest to us, UAV multicasting minimises the time for every ground terminal to receive a common file *directly* from the UAV [20]; UAV-aided sensor networks study the uplink [5]. We instead let terminals complete the file through a multi-hop mesh, and only a known subset needs it.

Dissemination and coding. Deluge [21] and Trickle [6] spread an object to every node by advertise–request epidemics, without knowing where data enters. Fountain codes [22], [7] and network coding [8] relax *which* packets are needed but not *where* they are; River is compatible with both.

Path planning. Boustrophedon coverage [9], [10] sweeps a region with parallel lanes; Dubins paths [11] make them flyable; the Dubins TSP with neighbourhoods visits a disc around each target [12]. Our path objective is the ground cooperation cost, not coverage.

A2G channels. Low-altitude A2G propagation depends strongly on environment and elevation angle [13], [14]; we therefore treat the range r as a regime parameter rather than a constant.

Person search at the edge. Face recognition [15], [16] and person re-identification [17] match a probe against a gallery of person crops by cosine similarity of learned embeddings, produced by a detector that crops each person; open-world settings must control false matches [18]. Embeddings of different models are incompatible [19]. Commercial cameras increasingly ship with fixed-function analytics, but legacy devices offer at most motion events [24]. We use these results only to define the query and the value of partial data.

III. SYSTEM MODEL

A. Network, UAVs and reception map

Network. The area is tiled by hexagonal cells (radius 100 m); nodes form an 802.15.4 mesh with one cell leader (CL) per cell and precomputed cell-level routes with one gateway pair between adjacent cells and a 7-colour channel reuse [23]. Clusters of cells each have a cluster head (CH) with a LoRa uplink to the BS. A fraction ϕ of cells host an AI node (a smart camera or an edge box); these are the *critical cells* $\mathcal{K}$. Other cameras are *legacy*: they detect motion by frame differencing [24] and keep event crops (≈15 kB each) in a local ring buffer.

UAVs. Team 1 (fixed-wing, 50 m/s, 100 m altitude, turn radius $\rho = 255$ m) broadcasts one 99 B payload every 10 ms, i.e., every $\Delta = 0.5$ m. Team 2 (rotary-wing, 15 m/s) verifies candidate cells with the full reference (≈78 s per cell).

Reception map. The packets emitted over cluster k are $D_k = \{s : \gamma(d_s) \in \mathcal{C}_k\}$; those emitted above cell a form $\mathrm{Seg}(a)$. After the pass node v holds $\mathrm{rx}(v) \subseteq D_k$, whose law depends on the distance from v to γ and on the A2G regime, summarised by the range r. The BS predicts this map before take-off.

B. Communication model

A2G link. At slant distance d from the UAV (altitude $h = 100$ m) the path loss at 2.4 GHz is free space up to $d_0 = 100$ m, $PL_0 = 80.05$ dB, and grows with exponent α beyond:

$$PL(d) = PL_0 + 10\alpha \log_{10}(d/d_0) + L_{\mathrm{obs}}, \quad d \geq d_0, \quad (1)$$

where L_{obs} is the excess loss of the environment (0 dB on open land, tens of dB under canopy or among buildings). Each packet sees independent Rician fading ($K = 2$) and is received if $P_t - PL(d) + X \geq S_{\min}$, with $P_t = +10$ dBm and $S_{\min} = -100$ dBm. We summarise the regime by the *range* r, the slant distance at which the mean received power equals $S_{\min}$:

$$r = d_0\, 10^{(P_t - PL_0 - L_{\mathrm{obs}} - S_{\min})/(10\alpha)}. \quad (2)$$

With $\alpha = 3$, $r \approx 1$ km for $L_{\mathrm{obs}} = 0$ and $r \approx 200$ m for $L_{\mathrm{obs}} \approx 21$ dB; sweeping L_{obs} reproduces the regimes of Section I. A node at lateral offset x hears the UAV over a flight stretch of $2\sqrt{r^2 - h^2 - x^2}$.

G2G link. Ground links follow exponent 3.5 with log-normal shadowing ($\sigma = 7.8$ dB) drawn once per node pair and Rayleigh fading in 100 ms blocks [28]; a shadowed pair stays shadowed. One acknowledged 802.15.4 hop (CSMA/CA, 127 B frame of 4.256 ms, ACK) takes $\tau_f \approx 5$–7 ms.

Data volume and flight time. Team 1 sends a $B = 99$ B payload every $T_p = 10$ ms, i.e., every $\Delta = vT_p = 0.5$ m. One copy of a query of S bytes is emitted over

$$L_{\mathrm{rot}} = \frac{vS}{R_{\mathrm{PHY}}\,\eta\,\delta} = Q\Delta \approx 2 \text{ km}, \quad (3)$$

with $R_{\mathrm{PHY}} = 250$ kb/s, payload efficiency $\eta = 8B/(T_p R_{\mathrm{PHY}}) = 0.317$ and duty cycle $\delta = 1$ at 2.4 GHz (sub-GHz bands impose $\delta \leq 0.1$–10% [25]). A cluster whose path has in-cluster length ℓ_k receives $|D_k| = \ell_k/\Delta$ packets and needs $\ell_k \geq Q\Delta$ to be offered one full copy; otherwise the UAV must extend the path or fly at $v \leq \ell_k/(QT_p)$. Team 1 needs $T_1 = (\sum_k \ell_k + \sum \ell_{\mathrm{link}})/v$, with Dubins links of radius ρ. The ratio $L_{\mathrm{rot}}/2r$ separates a regime in which nodes near the path hear a full copy ($\ll 1$) from one in which cooperation is mandatory ($\gg 1$).

Feedback and verification. A CH sends one LoRa message per $G = t_{\mathrm{air}}/\delta_{\mathrm{LoRa}} = 0.4/0.01 = 40$ s [25]. Team 2 verifies a cell in $\tau_v = \tau_{\mathrm{settle}} + N_{\mathrm{ref}}\tau_u + \tau_m \approx 5 + 10{,}592 \times 6.88\,\mathrm{ms} + 0.2 \approx 78$ s for a 1 MB reference sent by acknowledged unicast.

C. Query and matching abstraction

We make no image-processing contribution; this layer only states what the network must deliver and what partial delivery costs.

Query. The BS extracts $L = 12$ reference images of the target (2 face, 4 full body, 6 part images such as clothes or bag; ≈400 kB, $Q = 4{,}036$ packets). Each image is atomic: it is usable only when all its packets are present. Images rather than feature vectors are sent because AI nodes run different embedding models whose vectors are not comparable [19]; each node embeds the images with its own model and compares them, by cosine similarity, with the person crops it has stored [16], [17]. Q is a scenario parameter, swept from a face-only query (≈50 kB) to a rich one (≈800 kB).

Self-normalised decision. Since score scales differ across models, each node sets its own threshold from the scores of the strangers in its gallery so that its false-alarm rate is a common F (e.g., 5%), much like a detector setting its threshold from the measured noise floor. It reports one flag bit and a 2-byte bitmap of the images it holds, H, through its CH over LoRa.

Value of partial data. With $k_v(H)$ the images of aspect $v \in \{\text{face, body, part}\}$ held, a target that passed the camera is flagged with probability

$$V(H) = 1 - \prod_{v} \left[1 - \nu_v \left(1 - (1 - h_v)^{k_v(H)}\right)\right], \qquad (4)$$

where h_v is the per-image hit rate at false-alarm rate F (published TAR@FAR operating points) and ν_v the probability that aspect v is visible in the target's crop (faces are often occluded). For example, with $(h_v) = (0.8, 0.3, 0.1)$ and $(\nu_v) = (0.4, 0.9, 0.9)$, $V = 0.89$ with all 12 images but 0.42 with the six part images only. [TODO: source h_v, ν_v.]

BS inference and Team 2. With prior π_c that the target passed cell c (from the last-seen point and walking speed), a flag updates it to $p_c = \pi_c V / [\pi_c V + (1 - \pi_c)F]$ and a "no" to $p_c = \pi_c(1 - V)/[\pi_c(1 - V) + (1 - \pi_c)(1 - F)]$. A cell with incomplete data that answers "no" cannot be ruled out (e.g., $p_c = 3\%$ vs 0.6% for $\pi_c = F = 5\%$), so missing images cost Team 2 extra visits of ≈78 s each. Team 2 takes off at $t = 0$, visits the cell maximising p_c per unit time, and the mission metric is T_{hit}, the time a cell that truly saw the target is verified.

Legacy cameras. A legacy camera detects new objects by frame differencing, crops the changed region (≈15 kB) and keeps it in a ring buffer with a timestamp; outdoor false triggers (foliage, light) are filtered by blob size, aspect ratio and persistence. Its event rate is λ per hour.

D. Delivery problem

Given γ, deliver to each critical cell k a required set $D_k^\star$ (one complete copy of the query, or $\geq Q(1 + \epsilon)$ coded packets) minimising $T = \max_k T_k$ and the number of MAC frames, under half-duplex radios, one transmitter per channel, per-hop ACK and static shadowing. A single radio bounds $T_k \geq |\text{Lack}(k)|\, \tau_f$, with $\tau_f \approx 5$–6 ms per acknowledged hop.

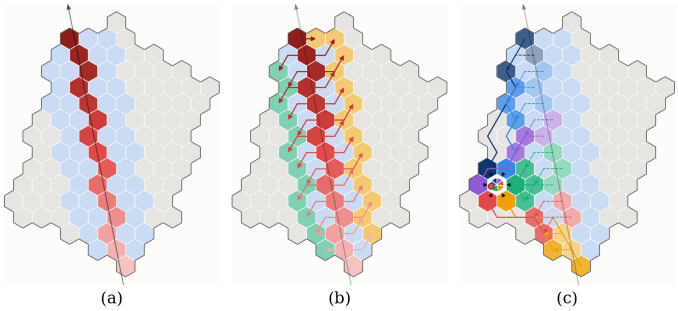


Fig. 1. River roles on a cluster with a straight path (arrow). (a) SPINE cells under the path (red, darker = later) and the River band (light blue). (b) Each SPINE cell serves one Bank on each side; arrows are the precomputed cell paths. (c) Ring of six helpers around a critical cell (red dot): each helper fetches one group of consecutive segments from the Banks and hands it over in one hop.

Algorithm 1 River planning at the BS (before take-off)

Input: path γ, cluster cells, critical cells $\mathcal{K}$, legacy cameras, r
SPINE $\leftarrow$ cells crossed by γ; $\text{Seg}(a) \leftarrow \{s : \gamma(d_s) \in a\}$
for all $a \in$ SPINE, side $\pm$ **do**
  walk along the local normal of γ up to ω or the edge
  $\text{Bank}_\pm(a) \leftarrow$ last cell reached; store the cell path
for all $k \in \mathcal{K}$ **do**
  $D_k^\star \leftarrow$ copy of the query emitted nearest to k ▷ window
  $J \leftarrow \text{clamp}(\lceil \widehat{\text{Lack}}(k)/\text{share} \rceil, 2, 12)$ helpers from rings 1–2
  split SPINE$\cap D_k^\star$ into J consecutive groups; helper j takes group j from the nearer Banks
for all legacy camera u **do** attach u to the nearest k by ETX; cap tree size and depth
send all tables to the CHs with the flight plan

Algorithm 2 River execution (after the pass)

Gather: in role cells only, nodes send to the CL what its role needs
Bank fill: Bank sends a "have" manifest of $\text{Seg}(a)$; SPINE cell a returns what is missing
Ring fill: helper j requests its group from its Banks
repeat final hop: k sends each helper its lack within the group; helper forwards
until k holds $D_k^\star$ (coded: $\geq Q(1 + \epsilon)$ packets)
Match and report flag $+H$ via CH
Event pull (per tree, frame budget B):
  periodically, or if a child's event rate exceeds $\lambda_{\max}$: pull its newest crops
  if k finds a suspicious candidate at t_d: pull from every child u the crops in $t_d \pm d(u, k)/v_{\text{walk}}$
All hops: ACK, retransmission, one transmitter per channel

IV. RIVER: PRE-PLANNED BIDIRECTIONAL TRANSPORT

Roles (Fig. 1). SPINE cells lie under the path and, after in-cell gathering, hold their own segment $\text{Seg}(a)$ almost surely, for any r. The River is the band of cells within ω of γ; for each SPINE cell, one *Bank* lies at distance $\approx \omega$ on each side. Around each critical cell, a *Ring* of up to six helpers splits the SPINE into consecutive groups, one per helper.

Planning (BS, before take-off; Algorithm 1). (i) SPINETABLE: $\text{Seg}(a)$ for every SPINE cell. (ii) Banks along the local normal of γ (curved paths included). (iii) Critical-cell windows: each critical cell takes the copy of the query emitted nearest to it. (iv) Rings, sized to the predicted shortfall (2–12 helpers). All tables are sent to the CH with

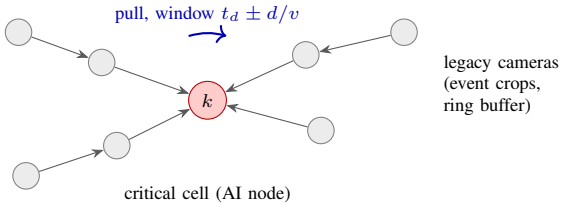


Fig. 2. Event-pull tree. The BS attaches each legacy camera to the nearest critical cell along precomputed routes. The critical cell pulls event crops periodically, when a child's event rate rises, and from the whole tree once it finds a suspicious candidate, restricted to a time window that grows with distance.

the plan.

Execution (after the pass; Algorithm 2). Gather to the CL inside role cells only; SPINE $\rightarrow$ Bank fill; Bank $\rightarrow$ helper fill (immediately, or on demand); helper $\rightarrow$ critical cell. Requests are "have" manifests in one frame; one transmitter per channel, round-robin over radios.

Why it is fast. Direct pulling sends every missing packet from the SPINE along one long path per critical cell, and these paths overlap. River replaces them with many short, parallel Bank $\rightarrow$ helper runs and one final hop. Because roles depend only on γ, River needs *no* route or content discovery.

Why it is robust to r. River sources data only from the SPINE, which receives its own segment whatever the A2G range. When r is large it simply has less to move; when r is small it still completes, whereas schemes relying on what a cell hears directly do not.

Coding. With a fountain-coded broadcast [22], River is unchanged except that manifests become counts: a helper forwards *any* coded packets of its group and the critical cell stops at $Q(1+\epsilon)$. Coding removes segment bookkeeping; River still supplies the packets a distant cell never heard.

Reverse direction: event pull. As shown in Fig. 2, each legacy camera is attached by the BS to the nearest critical cell along precomputed routes, forming a small tree. A critical cell pulls event crops from its tree (a) periodically or when a child's event rate exceeds a threshold, bounded by a frame budget, and (b) from the whole tree as soon as its matcher finds a suspicious candidate at time t_d, restricted to the window $t_d \pm d/v_{\text{walk}}$ for a child at distance d. The load of a tree is $n\,\lambda\,W\,b$ for n children, event rate λ, window W and crop size b; the BS bounds tree size and depth. The same planned routes thus serve both directions.

V. FLIGHT DESIGN: LM-TOUR

The path decides the reception map, so it is designed for River. The idea is simple: *visit a well-spread set of the most valuable AI nodes, and let River serve everything within a distance R of them.*

Node value. Each AI node u receives a weight $w(u)$ that summarises why visiting it closely pays off: its compute capability, the size of its legacy-camera tree, its prior π_u, and the River time it saves when the path passes near it rather than far away. The last term is predicted by a supervised surrogate

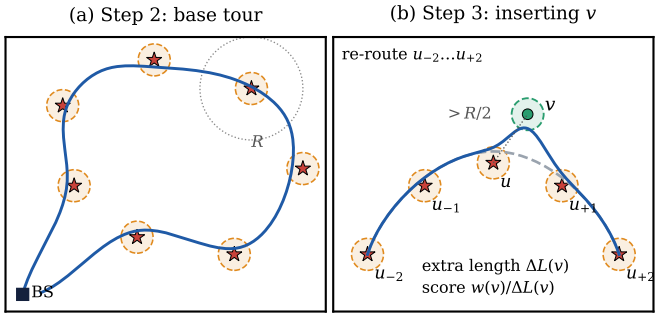


Fig. 3. LM-TOUR. (a) Step 2: landmarks (stars) are at least R apart; the Dubins tour from the BS passes through a disc of radius ξ around each. (b) Step 3: a candidate v farther than $R/2$ from its nearest landmark u is inserted by re-routing only the local stretch $u_{-2}, \ldots, u_{+2}$ (dashed: old, solid: new); the extra Dubins length $\Delta L(v)$ sets its score $w(v)/\Delta L(v)$.

trained on ns-3 runs (inputs: hops to the path, predicted shortfall, shared flows, A2G range r); the weights are inputs to the planner. Nodes sorted by decreasing w form the list $A^\star$.

Step 1: weighted R-net. Scan $A^\star$ and keep a node if it is at least R (graph distance over land) from every node already kept; the kept set $\mathcal{K}_1$ covers every AI node within distance R. Clusters are the graph-Voronoi regions of $\mathcal{K}_1$, grown by the capacity-limited multi-source Dijkstra GROW of [26] with the River load (tree size plus shortfall) as volume, so clusters are balanced.

Step 2: base tour. A Dubins tour (turn radius ρ) from the BS must pass within ξ of every node of $\mathcal{K}_1$ (Dubins TSP with neighbourhoods [12]; Fig. 3a). The visiting order comes from a TSP heuristic improved by 2-opt (reversing a tour segment whenever this shortens the tour, which removes self-crossings). For each disc we sample a few entry points and headings and pick, by dynamic programming over the fixed order, the combination with the shortest total Dubins length. The disc radius is

$$\xi = \min\big(\omega,\ \rho,\ \sqrt{r^2 - h^2}\big)\ : \tag{5}$$

within ω the node lies in the River band; $\xi \le \rho$ still lets the path turn by up to about 120° at a landmark, because an arc of radius ρ tangent to both legs of a turn θ stays $\rho(\sec(\theta/2) - 1)$ from the corner; and within $\sqrt{r^2 - h^2}$ the node hears the UAV directly. This gives $\xi \approx 255\,\text{m}$ at $r = 1\,\text{km}$ and $\approx 170\,\text{m}$ at $r = 200\,\text{m}$, well below $R/2$, which only sets the spacing of landmarks. Sharper turns are not forbidden; they are penalised by their Dubins length. [TODO: sweep $\xi \in \{100, 175, 250\}$ m: tour length and number of loops.]

Step 3: valuable insertions. A remaining node $v \in A^\star$ farther than $R/2$ from every visited node is a candidate. Its insertion re-routes the tour locally through the discs of its nearest visited node u and the two visited nodes before and after it (Fig. 3b) so that the path enters $B(v, \xi)$; the extra Dubins length is $\Delta L(v)$. Nodes are inserted greedily by score $w(v)/\Delta L(v)$ while the score exceeds a threshold θ and the tour fits the endurance $L_{\max}$.

Algorithm 3 LM-TOUR (v5: discs $B(\cdot,\xi)$)

Require: AI nodes with weights w, land graph G, R, ρ, ξ, θ, $L_{\max}$

$A^\star \leftarrow$ AI nodes sorted by w (descending); $\mathcal{K}_1 \leftarrow \emptyset$
for all $u \in A^\star$ **do**
  if $d_G(u, \mathcal{K}_1) \geq R$ **then** $\mathcal{K}_1 \leftarrow \mathcal{K}_1 \cup \{u\}$
clusters $\leftarrow$ GROW$(G, \mathcal{K}_1)$ ▷ balanced Voronoi regions
$\gamma \leftarrow$ Dubins TSP-N through discs $B(u,\xi)$, $u \in \mathcal{K}_1$
$\mathcal{C} \leftarrow \{v \in A^\star \setminus \mathcal{K}_1 : d(v, \text{visited}) > R/2\}$
repeat
  for all $v \in \mathcal{C}$ **do** $\Delta L(v) \leftarrow$ cheapest local re-route of γ into $B(v,\xi)$ passing the window check
  $v^\star \leftarrow \arg\max_v w(v)/\Delta L(v)$
  if $w(v^\star)/\Delta L(v^\star) < \theta$ **or** $|\gamma| + \Delta L(v^\star) > L_{\max}$ **then break**
  insert $v^\star$; remove from $\mathcal{C}$ all nodes within $R/2$ of $v^\star$
until $\mathcal{C} = \emptyset$
return γ, clusters, visited nodes

Window check. Every visited node must be offered a full copy of the query nearby: the stretch of $Q\Delta \approx 2\,\text{km}$ of path around its closest approach must lie within the River reach R_Q of the node ($R_Q \approx 1\,\text{km}$ for one window, from the saturation sweep). Candidates that break it are rejected or the stretch is straightened.

Choosing R. Kept nodes are pairwise at least R apart, so discs of radius $R/2$ around them are disjoint and $|\mathcal{K}_1| \leq 4A/(\pi R^2)$ (plus a boundary term) for an area A. A tour through n points in area A is at most $\sqrt{2nA} + O(\sqrt{A})$ [27], and Dubins turns add $O(n\rho)$, so the base tour grows as $1/R$, while River's delivery distance grows as R. Since $\xi \leq \rho$ and $\xi \leq \sqrt{r^2 - h^2}$, every landmark hears the UAV directly; we use $\xi = 255$, 255, 199 and 99 m at $r = 1000$, 500, 200 and 150 m, and sweep $\xi \in \{100, 175, 250\}$ m.

V. EVALUATION

Setup. ns-3.46 with `lr-wpan` (CSMA/CA, ACK, retransmission); channel and radio as in Section III-B ($\alpha = 3$). Three maps with 8, 12 and 15 clusters (45, 66 and 92 critical cells; capacitated k-medoids, at most 80 cells each). Four A2G regimes: $L_{\text{obs}} = 0$, 9.0, 19.8 and 25.7 dB in (1), calibrated on the ns-3 pass so that $\sqrt{r^2 - h^2}$ is 1015, 498, 199 and 99 m, i.e. $r \approx 1000$, 500, 200 and 150 m. Team 1 flies a speed profile (45° bank, $\pm 2\,\text{m/s}^2$), slowing to 25 m/s or looping where a cluster would not be offered one copy of the query (Eq. (3)); the clue set is fountain-coded and systematic. One ns-3 run per cluster and 100 paired draws of the matching layer per map: every scheme sees the same pass bitmaps, coded receptions and draws. Results are exploratory ($N < 120$).

Schemes. LM-Tour + River-FC (proposed, Section V). *CHORD + River-FC*: the longest flyable chord through each cluster head, bearings and visiting order chosen jointly to minimise the mean time critical cells are left behind. *FC + gather*: team 1 flies a TSP over the critical cells and circles each until its nodes hold enough coded packets; in-cell gathering only. *FC + DTSP-N*: as FC + gather, but nearby critical cells are grouped and served from orbits chosen by a sampled Dubins TSP with neighbourhoods. *Single pass, no cooperation*: CHORD's flight; each critical cell uses only what it decodes itself. *No broadcast (rotary TSP)*: team 2 alone visits every critical cell with the full reference. *Oracle*: team 2 flies straight to the target cell.

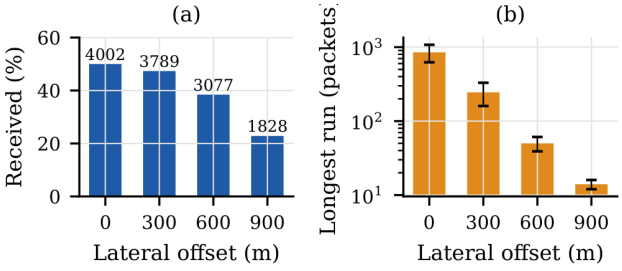


Fig. 4. A2G reception at $r \approx 1\,\text{km}$ (200 passes). (a) Packets received out of 8,001. (b) Longest error-free run. Reception decays slowly with lateral offset; contiguity collapses.

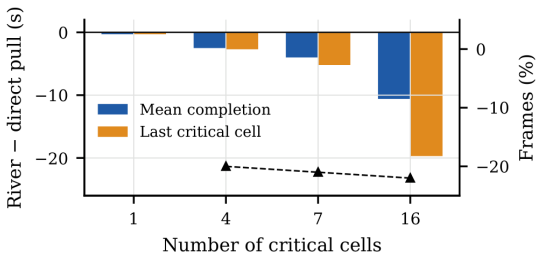


Fig. 5. River minus direct pull on a 91-cell cluster (20 paired passes): completion time (bars) and MAC frames (triangles) versus the number of critical cells.

A. Why cooperation is needed

Even in the favourable regime, a node 900 m from the path receives 1,828 of 8,001 packets but its longest run is 14 packets, against ≈250 for one face image (Fig. 4). Under short range a node hears only its own overhead segment.

B. River versus direct pulling

With 4, 7 and 16 critical cells River uses 20–22 % fewer frames in 20/20 passes and finishes the last critical cell 2.7, 5.2 and 19.7 s earlier (Fig. 5); with one critical cell on the SPINE the two coincide. On cluster C3 (80 cells, four critical cells) River completes the last critical cell in 37.2 s versus 50.3 s (−26 %) with 89.7 k versus 102.2 k frames (−12 %). At mission level the two give the same $T^\star$ within 3 % in 11 of 12 settings; River never stalls, while direct pulling leaves 6 of 92 critical cells incomplete at $r = 150\,\text{m}$.

C. Robustness to the A2G regime

Table I: without cooperation every critical cell completes at $r = 1\,\text{km}$, 66 % at 500 m and almost none at $r \leq 200\,\text{m}$. FC-based schemes complete every critical cell only by circling: the last one completes 2.1–5.2× later than with River-FC as r shrinks, whereas River-FC's completion grows by only 150–350 s, because River sources data from the cells under the path, which hear their own segment at any r.

TABLE I

CRITICAL CELLS COMPLETE AFTER THE PASS (%) AND TIME FROM TAKE-OFF TO THE LAST COMPLETION (S) VERSUS A2G RANGE, $K = 16$ (92 CRITICAL CELLS). [EST]: ESTIMATED.

Scheme	$r = 150$	200	500	1000 m
No cooperation	0 % / –	1 % / –	66 % / –	100 % / –
FC + gather	100 % / 7257	100 % / 6591	100 % / 2784	100 % / 1132
FC + DTSP-N	100 % / 6761	100 % / 5523	100 % / 2529	100 % / 1041
River-FC (CHORD)	100 % / 1395	100 % / 1398	100 % / 1204	100 % / 1052
River-FC + LM-Tour	100 % / 1283	100 % / 1268	100 % / 1082	100 % / 1018

TABLE II

MEAN MISSION TIME $T^\star$ (S), 100 PAIRED DRAWS, K = 8 / 12 / 16 CLUSTERS. [EST]: ESTIMATED, NOT RUN.

Scheme	$r = 1000$ m	500	200	150
LM-Tour + River-FC	647 / 686 / 863	775 / 942 / 1097	822 / 936 / 1143	810 / 942 / 1157
CHORD + River-FC	647 / 686 / 863	784 / 950 / 1107	876 / 975 / 1192	872 / 990 / 1215
FC + gather	660 / 815 / 949	1138 / 1451 / 1801	2285 / 3288 / 3964	2463 / 3460 / 4368
FC + DTSP-N	708 / 805 / 967	1078 / 1388 / 1750	1904 / 2482 / 3469	2190 / 3248 / 4468
Single pass, no cooperation	650 / 687 / 862	1379 / 2030 / 2549	3711 / 5277 / 6298	3860 / 5308 / 6358
No broadcast (rotary TSP)	2926 / 4168 / 5169	2926 / 4168 / 5169	2926 / 4168 / 5169	2926 / 4168 / 5169
Oracle	322 / 402 / 404	322 / 402 / 404	322 / 402 / 404	322 / 402 / 404

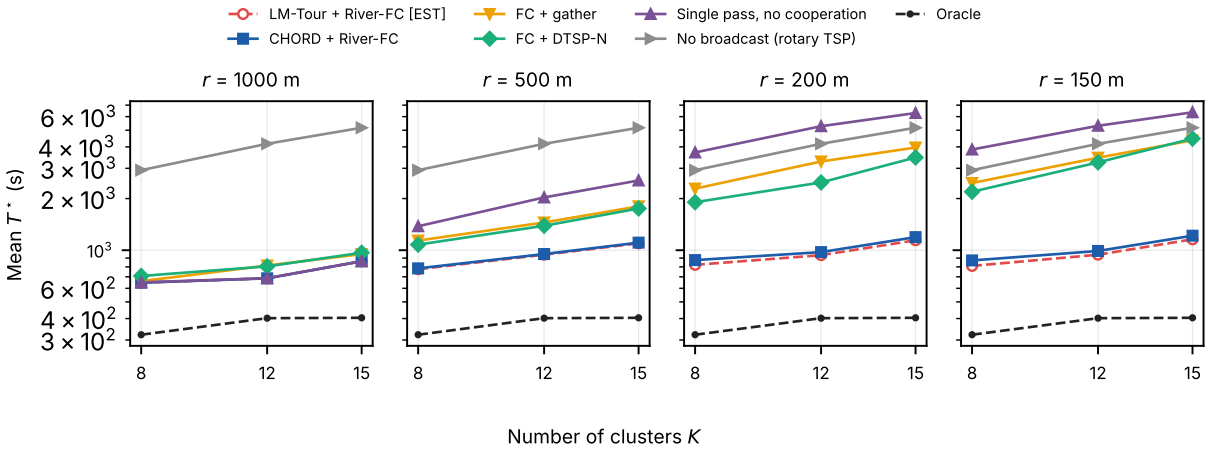


Fig. 6. Mean $T^\star$ versus the number of clusters, per A2G range (log scale). LM-Tour estimated [EST]. Oracle knows the target cell.

D. Mission level

Table II, Fig. 6: the single-pass architecture with River-FC is 3.3–6.0× faster than having no broadcast phase. Against the FC baselines it is 2–15 % faster at $r = 1$ km and 27–72 % faster at $r \leq 500$ m, and the gap widens with K: CHORD's flight stays 782 / 1152 / 1415 s at every r, while FC's grows 2–6×. Without cooperation the same flight is 1.8–5.4× slower at $r \leq 500$ m. [EST] LM-Tour shortens $T^\star$ by a further 1–7 % at $r \leq 500$ m (no change at 1 km), by bringing the path within direct hearing of each landmark.

VI. CONCLUSION

In a single-pass UAV broadcast the flight path is the transmission schedule, so the BS can plan the ground's cooperation before take-off. River turns this into a pre-planned, bidirectional transport that delivers the query to the few AI-capable cameras regardless of A2G range, works with fountain coding, and pulls event frames from legacy cameras over the same routes. Against UAV-served fountain-coded baselines the mission is 27–72 % faster once the A2G range falls to 500 m or below. [EST] LM-Tour adds 1–7 %. [TODO: field A2G measurements.]

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